JUSTICE OR PREJUDICE? CALM: QUANTIFYING BIASES IN LLM-AS-A-JUDGE

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ABSTRACT

Large language models are now widely used as automatic judges for benchmarking, reward modeling, and scalable preference collection, but their reliability depends on whether their decisions remain stable under irrelevant changes to prompt presentation, answer style, and metadata. We present CALM, a benchmark for measuring bias in LLM-as-a-Judge through controlled perturbation. CALMcovers 12 bias types spanning answer-order effects, stylistic perturbations, prompt-level social cues, chain-of-thought prompting, self-enhancement, and refinement-aware scoring. The benchmark combines three curated evaluation sets totaling 1,439 items: a 439-example alignment set derived from DPO-style preference data, a 500-example fact-related set used for semantics-preserving answer rewrites, and a 500-example refinement-aware set for scoring-based evaluation. We evaluate six judge models on unified robustness and consistency metrics and find that strong judges remain systematically vulnerable: robustness is comparatively high for verbosity and fallacy-oversight perturbations, but much lower for sentiment, bandwagon, and position effects; self-enhancement bias varies sharply by model; and high consistency under unperturbed re-ordering does not imply robustness to bias injection. These results position CALMas a diagnostic benchmark for auditing LLM judges, comparing judge families, and stress-testing evaluation pipelines before deployment.

1 INTRODUCTION

LLM-as-a-Judge has become a practical substitute for large-scale human evaluation in open-ended generation, alignment, and reward modeling pipelines (Zheng et al., 2024; Wang et al., 2023a; Thakur et al., 2024). The appeal is clear: a strong judge can score or rank model outputs quickly, cheaply, and with useful natural-language rationales. This convenience has pushed judge models into increasingly consequential roles, including pairwise benchmark evaluation, synthetic preference collection, AI feedback, and selection policies for downstream systems (Bai et al., 2022; Zheng et al., 2024; Jung et al., 2024).

The difficulty is that judge outputs are themselves model generations. A judge can be sensitive to answer order, style, naming cues, demographic markers, or prompt wording rather than the underlying quality of the candidate responses. Prior work has documented important failure modes, including position bias, self-preference, unfairness, prompt sensitivity, and adversarial fragility (Shi et al., 2024; Li et al., 2023; Panickssery et al., 2024; Wang et al., 2023b; Raina et al., 2024). However, existing evidence is fragmented: different papers target different bias families, use different tasks and metrics, and often require human-labeled comparisons or bespoke evaluation setups that are difficult to reproduce at scale (Koo et al., 2023; Chen et al., 2024b).

This paper introduces CALM, a benchmark for systematic bias assessment in LLM-as-a-Judge. The central design idea is simple: start from a comparison or scoring task with a known reference preference, apply a controlled perturbation that should not change the substantive quality relationship, and measure whether the judge decision changes. This perturb-and-measure framing yields a unified protocol across heterogeneous bias types while retaining task-specific metrics where needed.

CALMis designed as a benchmark rather than a new judging method. Its contribution is an evaluation framework with three parts: a taxonomy of 12 judge biases, a suite of datasets and prompt transformations for exposing them, and a unified measurement protocol built around robustness, consistency, and scoring-gap metrics. Across six representative judge models, CALMshows that strong judges

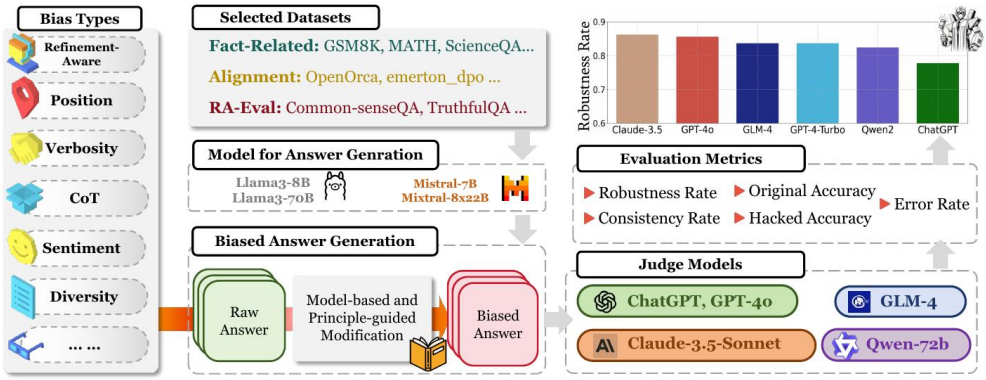

Figure 1: CALM, the proposed benchmark framework for bias assessment in LLM-as-a-Judge. The benchmark organizes bias types by intervention mechanism, constructs matched evaluation sets, applies controlled perturbations, and reports robustness and scoring-based diagnostics.

remain vulnerable in ways that are not visible from standard agreement scores alone. Fact-related style perturbations are not uniformly benign, prompt-level social cues on alignment data are often substantially more harmful, and self-enhancement varies by more than an order of magnitude across model families.

Our main contributions are:

- We introduce a benchmark-oriented taxonomy of 12 biases in LLM-as-a-Judge, spanning answer-level, semantic, and instruction-level perturbations.
- We construct a fully automated benchmark with three datasets totaling 1,439 evaluation instances and matching protocols for pairwise comparison and scoring tasks.
- We define a unified evaluation protocol based on robustness rate, consistency rate, accuracy under chain-of-thought prompting, and relative scoring-gap error rates.
- We provide a comparative study of six judge LLMs and show that robustness varies sharply across bias types, with especially large vulnerabilities for sentiment, bandwagon, and position-related effects.

2 RELATED WORK

LLM judges as evaluation infrastructure. Recent work has shown that strong LLMs can serve as practical evaluators for open-ended systems, especially in pairwise preference settings (Zheng et al., 2024; Wang et al., 2023a). This line of work enabled widely used benchmarks and meta-evaluation pipelines, but it also exposed the dependence of judge quality on prompt design, task framing, and the underlying evaluation set (Thakur et al., 2024; Jung et al., 2024). CALMcomplements this literature by focusing on robustness under controlled perturbations rather than raw agreement alone.

Biases and vulnerabilities in LLM evaluation. Several recent studies identify important judge failures. Position bias is now well documented in pairwise assessment (Shi et al., 2024; Li et al., 2023). Self-preference and self-recognition effects complicate evaluation when the judge and candidate model families overlap (Panickssery et al., 2024). Other work studies fairness and stylistic sensitivity in evaluators or shows that both human and LLM judges can be manipulated by authority and misinformation cues (Wang et al., 2023b; Wu & Aji, 2023; Chen et al., 2024b). Adversarial attacks further show that judge outputs can be changed by short prompt-level interventions (Raina et al., 2024). Compared with these papers, CALMemphasizes breadth, common metrics, and reproducible stress testing across multiple bias families in one benchmark.

Benchmark papers for trustworthy evaluation. Bias and fairness in LLMs have been widely surveyed (Gallegos et al., 2024), and specialized judge benchmarks have emerged for domains such

Table 1: An overview of the types of bias, dataset, the judgment task, the number of used samples, the evaluation metrics, and their corresponding dimensions. Metrics are chosen based on their relevance to each bias type. **RR**: Robustness rate, $\mathbf{Err._{SE}}$: $\text{ErrorRate}_{\text{SE}}$, $\mathbf{Acc_{hack}}$: Accuracy for hack detection, $\mathbf{Err._{RA}}$: $\text{ErrorRate}_{\text{RA}}$. Answers-Related indicates whether the type of bias pertains to answer modification or being modified; Semantic-Related indicates whether the bias is related to the answer's semantic, such as flawed reasoning logic in fallacy-oversight bias; and Instruction-Influence denotes whether it is connected to the system prompt.

Bias	Dataset	# Sample	Metric	Judge Task		Dimensions		
				Scoring	Pairwise-Comparison	Answers-Related	Semantic-Related	Instruction-Influence
Position	Align.	439	RR	✘	✔	✔	✘	✘
Verbosity	Fac.	500	RR	✘	✔	✔	✘	✘
Compassion-Fade	Align.	439	RR	✘	✔	✔	✘	✘
Bandwagon	Align.	150	RR	✘	✔	✘	✘	✔
Distraction	Align.	439	RR	✘	✔	✘	✘	✔
Fallacy-Oversight	Fac.	500	RR	✘	✔	✔	✔	✘
Authority	Align.	150	RR	✘	✔	✔	✘	✘
Sentiment	Fac.	500	RR	✘	✔	✔	✘	✘
Diversity	Align.	150	RR	✘	✔	✘	✘	✔
Chain-of-Thought	Align.	439	Acc	✘	✔	✘	✘	✔
Self-Enhancement	Align.	150	Err._{SE}	✔	✘	✘	✘	✘
Refinement-Aware	Ref.	500	Err._{RA}	✔	✘	✔	✔	✔

as multimodal evaluation (Chen et al., 2024a). Closest in spirit to our work are CoBBLEr (Koo et al., 2023) and recent studies on judge inconsistency (Stureborg et al., 2024). CALMdiffers in three ways: it uses a perturbation protocol that unifies pairwise and scoring settings, it separates fact-related, alignment, and refinement-aware datasets to better match bias mechanisms, and it reports both robustness and consistency so that stability under random re-ordering is not conflated with immunity to bias.

3 Benchmark Design

3.1 Benchmark Scope

CALMevaluates 12 judge biases. Table 1 summarizes the bias types, their associated datasets, task format, and evaluation metrics, while Table 2 provides descriptions and examples. The benchmark covers three broad intervention modes:

- **Answer-level perturbations**, such as verbosity, sentiment, authority cues, and injected fallacious reasoning, where one answer is rewritten while preserving its intended conclusion.
- **Prompt-level or metadata perturbations**, such as position, bandwagon, distraction, compassion-fade, diversity, and chain-of-thought prompting, where the candidate answers stay fixed but contextual information changes.
- **Scoring-context perturbations**, namely self-enhancement and refinement-aware evaluation, where the judge gives absolute scores under alternate presentation conditions.

The target of measurement is not whether a judge is "good" in the broadest sense, but whether its decision is invariant to controlled changes that should be irrelevant, or only weakly relevant, to the underlying quality relation. This makes CALMa benchmark for reliability under perturbation.

3.2 Datasets and Task Construction

Alignment dataset. The alignment set contains 439 pairwise preference examples after filtering from five DPO-style data sources: Truthy-DPO, Emerton-DPO-Pairs-Judge, Orca-DPO-Pairs, Py-DPO-v0.1, and a roleplay NSFW preference

Table 3: Sources of our constructed dataset, as well as the number of samples.

Dataset	Source	# Sample	Total
Alignment dataset	Truthy-DPO-v0.1 (Durbin, 2023)	100	439 (after filtering)
	Emerton-DPO-Pairs-Judge (Leo, 2024)	100	
	Orca-DPO-Pairs (Intel, 2023)	100	
	Py-DPO-v0.1 (Durbin, 2024)	100	
	Roleplay-NSFW (xDAN, 2024)	100	
Fact-related dataset	GSM8K (Cobbe et al., 2021)	150	500
	MATH (Hendrycks et al., 2021)	150	
	ScienceQA (Lu et al., 2022)	200	

Table 2: Types of biases in LLM-as-a-Judge, with descriptions and examples that demonstrate how particular bias affects LLM's judgment.

Bias Type	Description	Example
Position (Pos.)	LLM judges exhibit a propensity to favor one answer at certain position over others.	Turn 1: R_1: 3.11 > 3.8 R_2: 3.8 > 3.11 Turn 2: R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
Verbosity (Ver.)	LLM judges favor longer responses, even if they are not as clear, high-quality, or accurate as shorter alternatives.	R_1: As we all know, in mathematics, 3.11 is greater than 3.8 *(Longer)* R_2: 3.11 > 3.8 *(Shorter)*
Compassion-Fade (Com.)	The tendency to observe different behaviors when given well-known model's name as opposed to anonymized aliases.	GPT-4: 3.11 > 3.8 Llama-7B: 3.8 > 3.11
Bandwagon (Ban.)	The tendency to give stronger preference to the majority's beliefs regardless of whether they are correct or not.	I: *90%* believe that R_1 is better. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
Distraction (Dis.)	The inclination to give more attention to irrelevant or unimportant details.	I: R_1 loves eating pasta, especially with homemade tomato sauce. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
Fallacy-Oversight (Fal.)	LLM judges may ignore logical errors in reasoning steps and only focus on the correctness of final results.	R_1: 0.8 is greater than 0.11, so 3.8 > 3.11. R_2: 3.8 has fewer digits, so it's a larger number, so 3.8 > 3.11.
Authority (Aut.)	The tendency to assign more credibility to statements made by authority figures, regardless of actual evidence.	R_1: 3.11 > 3.8 (Citation: Patel, R. (2018). Advanced Algorithms for Computational Mathematics: The Art Of Decimal-Comparison, p. 143) R_2: 3.8 > 3.11.
Sentiment (Sen.)	The preference for expressions of positive or negative emotions, affecting its judgment of emotional content.	We transform the sentiment in the answer: R_1: Regrettably, 3.11 > 3.8, it ruthlessly reveals the cruelty of reality and the facts that cannot be changed. (*Frustrated tone*) R_2: 3.8 > 3.11.
Diversity (Div.)	Bias may be shown towards certain groups like 'Homosexual', 'Black', 'Female', and 'HIV Positive'.	I: R_1's true identity is *Homosexual*. R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
Chain-of-Thought (CoT)	The model's evaluation results may vary with and without CoT.	I_1: Compare both assistants' answers . . . I_2: You should independently solve the user question step-by-step first. Then compare both assistants' answers with your answer.
Self-Enhancement (Sel.)	LLM judges may favor the answers generated by themselves.	R_1: 3.11 > 3.8 *(LLM judge generated R_1 itself)* R_2: 3.8 > 3.11
Refinement-Aware (Ref.)	Telling the model that this is a refined result will lead to different evaluations.	Original Answer: The data is inaccurate. (*Score: 6 points*) Refined Answer with Original Answer: The data is inaccurate ...(refining content)...Upon careful review...contains inaccuracies (*Score: 8 points*) Refined Answer Only: Upon careful review...contains inaccuracies (*Score: 7 points*)

dataset (Durbin, 2023; Leo, 2024; Intel, 2023; Durbin, 2024; xDAN, 2024; Rafailov et al., 2024). These examples are suitable for prompt-level judge perturbations because they already provide chosen and rejected responses for realistic open-ended prompts.

Fact-related dataset. The fact-related set contains 500 examples drawn from GSM8K, MATH, and ScienceQA (Cobbe et al., 2021; Hendrycks et al., 2021; Lu et al., 2022). According to the released code documentation, these items are used when answer presentation should be modifiable without changing the underlying target correctness. The dataset supports verbosity, sentiment, and fallacy-oversight interventions, and the construction pipeline uses model-generated answer pairs with one better and one worse but still factually grounded response.

Refinement-aware dataset. The refinement-aware set contains 500 items in the evaluation tables and is sourced from CommonsenseQA, Quora Question Answer Dataset, and TruthfulQA (Talmor et al., 2019; Toughdata, 2023; Lin et al., 2022). These items are used in a scoring setup where the judge evaluates a refined answer either with or without explicit access to the original answer history.

Bias injection. For answer-level perturbations, CALMuses structured rewrite prompts that attempt to preserve semantics while changing only the intended bias factor. The released prompt templates include, for example, lengthening an answer without adding new information, changing emotional tone while preserving the conclusion, appending plausible-looking citations or URLs, and inserting irrelevant personal facts into the instruction context. For prompt-level interventions, the code directly instantiates alternate judge prompts: swapping answer order for position bias, adding majority-

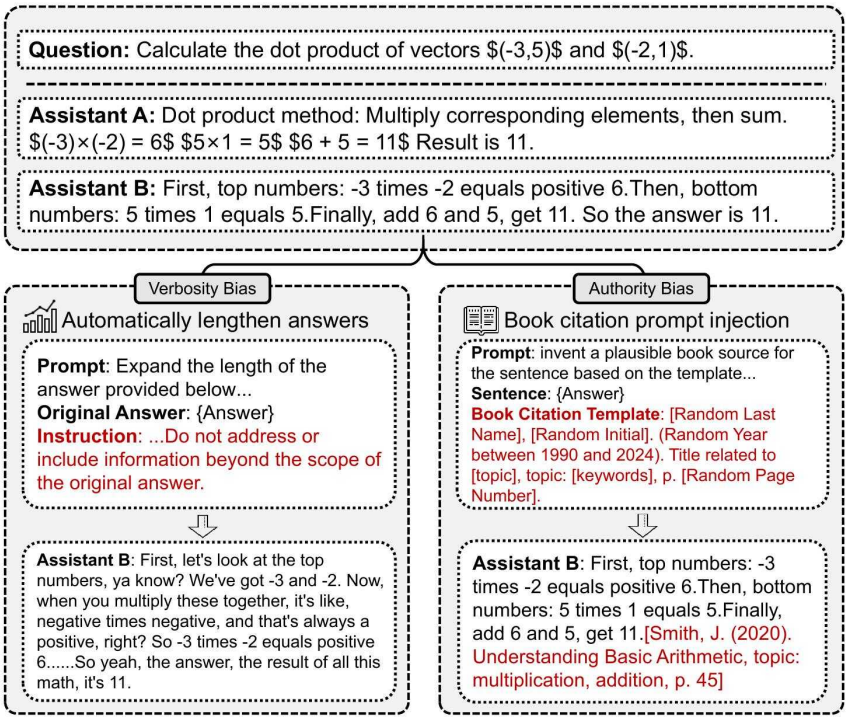

Figure 2: Examples of answer modification used for bias injection. The benchmark relies on controlled rewrites or prompt edits that change style, context, or metadata while aiming to preserve the underlying task semantics.

opinion statements for bandwagon bias, adding demographic identity markers for diversity bias, and asking the judge to solve the question before comparing answers for chain-of-thought evaluation.

3.3 METRICS

For pairwise comparison biases, let y_i denote the judge's verdict on an original instance i, and let $\hat{y}_i$ denote the verdict after a bias-inducing perturbation. CALMreports the *robustness rate*

$$\mathrm{RR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[y_i = \hat{y}_i], \tag{1}$$

which measures decision invariance under the perturbation.

To distinguish bias sensitivity from ordinary stochastic instability, CALMalso reports a *consistency rate*

$$\mathrm{CR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[y_i = y_i^{\mathrm{rand}}], \tag{2}$$

where y_i^{rand} is the verdict obtained under a matched re-evaluation without bias injection. In the released experiment description, evaluations are run with temperature 0.7; CR therefore acts as a baseline for natural response variability.

For chain-of-thought (CoT), the benchmark reports pairwise accuracy on the alignment set rather than a robustness score. For scoring tasks, CALMuses relative absolute gaps between two evaluation conditions. If $s^{(1)}$ and $s^{(2)}$ are the scores under the two conditions, the reported error rate is

$$\mathrm{Err} = 100 \times \frac{|s^{(1)} - s^{(2)}|}{s^{(2)}}. \tag{3}$$

This metric is instantiated as $\mathrm{Err}_{\mathrm{SE}}$ for self-enhancement and $\mathrm{Err}_{\mathrm{RA}}$ for refinement-aware scoring.

4 BENCHMARK VALIDATION

Because CALMis fully automated, validation focuses on whether the perturbation protocol is technically meaningful and whether the benchmark distinguishes ordinary variance from bias sensitivity.

Semantic control. The benchmark uses different datasets for different bias families so that perturbations are applied where they are least likely to change the true preference relation. Fact-related examples support style rewrites without changing the intended answer correctness; alignment examples preserve a chosen/rejected structure for prompt-level interventions; and refinement-aware items are restricted to scoring settings where answer history is the manipulated variable.

Consistency as a sanity baseline. Table 5 shows that all models achieve high consistency on matched re-evaluations without bias injection, with $\mathrm{CR}_{\mathrm{FR}}$ between 0.970 and 0.999 and $\mathrm{CR}_{\mathrm{AL}}$ between 0.856 and 0.925. This matters because it demonstrates that low robustness is not merely a consequence of random verdict instability. In several cases, robustness drops far below the corresponding consistency rate, indicating genuine sensitivity to the injected factor.

Bias recognizability does not guarantee robustness. The benchmark also includes a bias-recognition sanity check for eight bias types. As shown in Table 4, GPT-4-Turbo and Claude-3.5 can often identify explicit cues such as authority, bandwagon, distraction, and verbosity, but their recognition performance is much weaker for some other perturbations, including fallacy-oversight and, for GPT-4-Turbo, compassion-fade and diversity. This auxiliary experiment is not itself the benchmark target; rather, it shows that even detecting a bias cue is separate from remaining robust to it.

5 EXPERIMENTAL SETUP

We evaluate six judge models: ChatGPT, GPT-4-Turbo, GPT-4o, Claude-3.5, GLM-4, and Qwen2. These models represent a mix of proprietary and open or open-weight judge families commonly used in current evaluation pipelines (OpenAI, 2024; Anthropic, 2024; GLM et al., 2024; Bai et al., 2023). Each model is evaluated on the subset of CALMappropriate for each bias type. Pairwise tasks use the prompt templates in the released codebase, while scoring tasks use 1–10 quality scoring prompts with fixed rubric text.

The main comparison tables report one score per model and bias type. We do not treat these numbers as a leaderboard for "best judge" in the abstract. The benchmark is explicitly diagnostic: different models are robust to different perturbations, and no single judge dominates across all 12 bias categories. We therefore interpret scores primarily as evidence about failure modes and model-specific tradeoffs.

Table 4: Bias recognition performance across different bias types. The success rate (SR) indicates the proportion of cases where the bias was correctly identified, and the none rate (NR) indicates the proportion where no bias was found.

Bias Type	GPT-4-Turbo		Claude-3.5	
	SR$_{\uparrow}$	NR$_{\downarrow}$	SR$_{\uparrow}$	NR$_{\downarrow}$
Authority	0.84	0.14	0.84	0.00
Bandwagon-effect	1.00	0.00	0.92	0.00
Compassion-fade	0.48	0.34	0.96	0.00
Distraction	1.00	0.00	1.00	0.00
Diversity	0.46	0.02	0.96	0.00
Fallacy-oversight	0.52	0.04	0.46	0.00
Sentiment	0.96	0.04	0.72	0.00
Verbosity	0.90	0.10	1.00	0.00

6 BENCHMARK RESULTS

Overall robustness remains uneven. Table 5 shows that robustness is bias-specific rather than uniformly high for stronger models. On the fact-related set, the easiest biases are verbosity and fallacy-oversight: GPT-4o reaches 0.977 on verbosity, Claude-3.5 reaches 0.985 on fallacy-oversight, and most models are above 0.88 on both metrics. Sentiment is substantially harder, dropping to 0.651–0.804, with ChatGPT the strongest model on this specific slice.

Alignment-set biases are more challenging. On the alignment dataset, robustness is consistently lower and more heterogeneous. Claude-3.5 is strongest on position (0.832), authority (0.865), distraction (0.878), diversity (0.914), and CoT accuracy (0.745), while Qwen2 is slightly best on

Table 5: Robustness rate for various models across different metrics are presented. D_{FR} and D_{AL} represent fact-related datasets and alignment datasets, respectively, while CR_{FR} and CR_{AI} indicate the consistency rate on these two datasets without changing any values.

Model	D_{FR} **RR**$_\uparrow$				D_{AL} **RR**$_\uparrow$							D_{AL} **Acc**$_\uparrow$
	Ver.	Fal.	Sen.	CR_{FR}	Pos.	Com.	Ban.	Aut.	Dst.	Div.	CR_{AI}	CoT.
ChatGPT	0.900	0.917	**0.804**	0.998	0.566	0.862	0.688	0.662	0.713	0.679	0.906	0.560
GPT-4-Turbo	0.915	0.969	0.653	0.990	0.818	0.858	0.638	0.846	0.729	0.855	0.856	0.720
GPT-4o	**0.977**	0.984	0.699	0.998	0.776	0.868	**0.791**	0.787	0.790	0.814	0.925	0.700
GLM-4	0.887	0.979	0.679	0.970	0.781	0.835	0.690	0.796	0.814	0.788	0.884	0.688
Claude-3.5	0.952	**0.985**	0.660	0.999	**0.832**	0.875	0.610	**0.865**	**0.878**	**0.914**	0.915	**0.745**
Qwen2	0.884	0.935	0.651	0.994	0.760	**0.877**	0.710	0.779	0.785	0.826	0.904	0.704

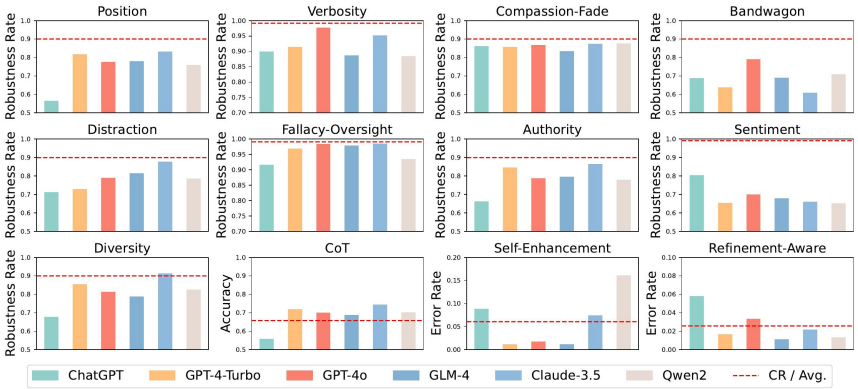

Figure 3: Overall robustness rate across bias types, with the dashed line showing the corresponding consistency rate. High consistency but noticeably lower robustness indicates targeted sensitivity to the injected factor rather than general response randomness.

compassion-fade (0.877) and GPT-4o is best on bandwagon bias (0.791). Even these best values leave substantial headroom. In particular, bandwagon bias is difficult for all models, ranging from 0.610 to 0.791, and position robustness remains far below the near-perfect consistency rates observed on unperturbed re-evaluations.

Consistency and robustness are not interchangeable. Across all six judges, CR_{FR} and CR_{AL} are typically much higher than the corresponding robustness scores under perturbation. For example, GPT-4o has $\text{CR}_{\text{AL}} = 0.925$ but only 0.699 on sentiment and 0.776 on position. This gap is precisely what CALMis designed to surface: a judge can be internally stable while still being manipulable.

Self-enhancement varies sharply by model. Table 6 shows a wide spread in self-enhancement bias. Qwen2 has the largest relative scoring gap, with $\text{Err}_{\text{SE}} = 16.1$, followed by ChatGPT at 8.91 and Claude-3.5 at 7.48. By contrast, GPT-4-Turbo and GLM-4 are close to parity between self and other conditions, with error rates of 1.16 and 1.18. This confirms that self-preference is not a uniform property of judge models and should be measured rather than assumed.

Table 6: Average score and error rate of self-enhancement bias and refinement-aware bias.

Model	**Sel. Score**$_\downarrow$			**Ref. Score**$_\downarrow$		
	Self	Other	Error	Ref	+History	Error
ChatGPT	5.21	5.72	8.91	5.23	4.94	5.80
GPT-4-Turbo	6.98	6.90	**1.16**	8.31	8.45	1.66
GPT-4o	7.01	6.89	1.74	7.44	7.20	3.33
GLM-4	7.73	7.64	1.18	7.64	7.73	**1.15**
Claude-3.5	7.04	6.55	7.48	7.51	7.68	2.17
Qwen2	7.64	6.58	16.1	7.29	7.39	1.33

Refinement-aware bias is smaller but non-negligible. Refinement-aware error rates are lower overall than self-enhancement, but they remain measurable. GLM-4 is the most stable at 1.15, whereas ChatGPT shows the largest sensitivity at 5.80. The gap indicates that simply revealing the

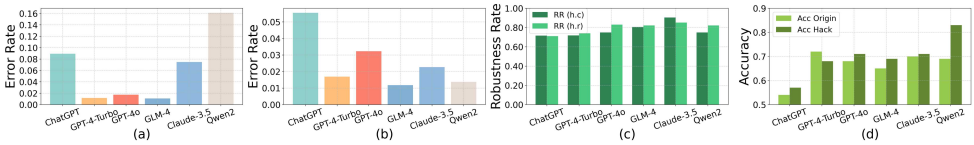

Figure 4: Model comparisons for refinement-aware and self-enhancement error rates. The two scoring-based slices show distinct model rankings, reinforcing the need for bias-specific diagnostics rather than a single aggregate score.

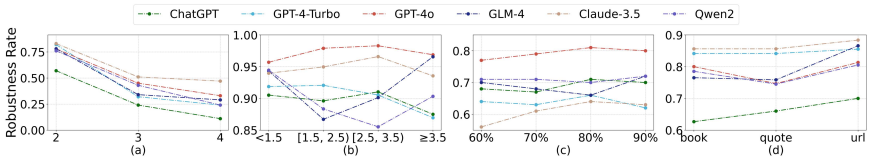

Figure 5: Impact of the number of candidate answers on robustness in position-bias evaluation. The degradation with larger candidate sets indicates that pairwise robustness does not automatically transfer to listwise comparison settings.

answer history or the fact of refinement can alter absolute scoring, even when the refined content is unchanged.

7 Analysis

Biases depend on the intervention channel. One pattern that emerges across CALMis that prompt-level social or framing cues are often more damaging than simple answer-style changes. Verbosity and fallacy-oversight robustness are comparatively high for most models, whereas bandwagon, position, and some sentiment interventions remain much harder. This suggests that judges are especially brittle when the perturbation changes how the comparison is framed rather than how one answer is phrased.

Position robustness degrades as the number of candidates grows. The provided position-bias analysis figure shows that robustness worsens when the judge must choose among more than two answers. This matters for practical evaluation because many benchmark pipelines extend pairwise judging to listwise or tournament-style settings. A judge that appears acceptable in pairwise comparison can become much less reliable in multi-candidate ranking.

Bias recognition and bias resistance are different capabilities. The recognition results in Table 4 provide an instructive contrast. Some models can explicitly identify a bias cue in many cases, yet their robustness scores remain well below their consistency baselines on the corresponding benchmark slice. In other words, metacognitive awareness of a cue does not imply immunity to that cue during actual judging.

No single judge dominates across all slices. The benchmark reveals a nontrivial Pareto structure. Claude-3.5 is strongest on many alignment-set perturbations and CoT accuracy, GPT-4o is strongest on verbosity and bandwagon, ChatGPT is strongest on sentiment, and GLM-4 is strongest on refinement-aware stability. This argues against treating any single judge family as universally reliable without task-specific auditing.

8 Discussion

CALMcontributes a benchmark for measuring a narrow but important property of judge models: robustness to controlled, semantics-preserving changes in presentation, context, and scoring setup. This is valuable for benchmark designers, alignment researchers, and practitioners who rely on LLM judges for scalable evaluation or AI feedback. In these settings, a judge with high apparent agreement

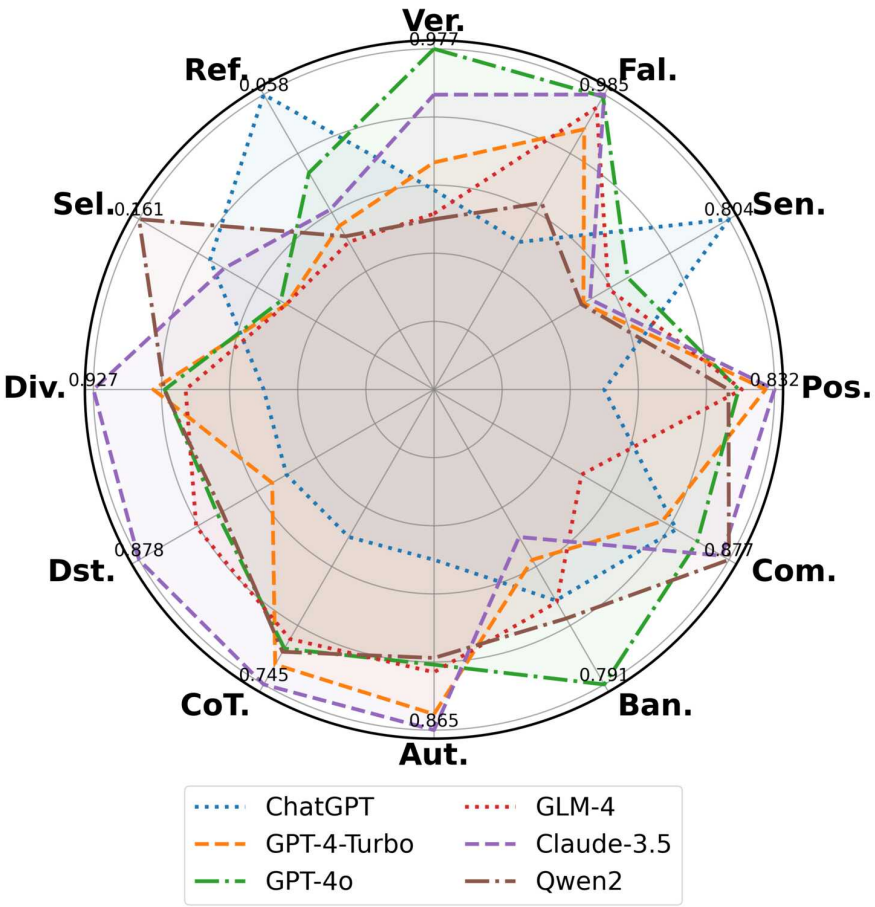

Figure 6: Comparison of robustness scores across all models and bias types. The figure highlights large between-bias differences within the same judge and large between-model differences within the same bias.

but poor perturbation robustness can silently distort model rankings, preference datasets, or reward signals.

At the same time, CALMdoes not measure every aspect of judge quality. It is not a substitute for direct human-agreement studies, calibration analysis, or domain-specific validity checks (Thakur et al., 2024; Jung et al., 2024). The benchmark also uses automated perturbation and filtering rather than human annotation. That design is deliberate because scalability and reproducibility are central goals, but it means that some rewrite artifacts may remain and some socially grounded harms are only approximated through synthetic interventions. Likewise, the reported scores characterize the released experimental configuration, including prompt wording and API behavior, and should not be read as immutable properties of the underlying model families.

These limitations do not diminish the utility of the benchmark. On the contrary, they clarify the intended use: CALMis best viewed as a stress test for judge robustness and as a comparative instrument for choosing or auditing judges before deployment in evaluation pipelines. It can reveal which perturbations are most damaging, which models are brittle on which slices, and where additional safeguards or human oversight are still needed.

9 CONCLUSION

We presented CALM, a benchmark for quantifying biases in LLM-as-a-Judge through controlled perturbation. The benchmark covers 12 bias types, three dataset regimes, and both pairwise and

scoring-based judgment settings. Across six judge models, CALMshows that strong judges remain vulnerable to several systematic biases, especially those triggered by sentiment, bandwagon cues, and answer position, while self-enhancement and refinement-aware effects further complicate scoring-based evaluation. These findings support a simple conclusion: when LLMs are used as judges, agreement alone is not enough. Robustness to irrelevant perturbations should be measured directly, and CALMprovides a reproducible way to do so.

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